$\mathscr{T}$: The Chronological "Operator"

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UO

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What is $\mathscr{T}$? A few different names...

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The Chronological Operator

> "...reorders its argument such that the times are in increasing order from right to left." -Steck, QM

The Question: What is it?

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$$T\{AB\} = \begin{cases} (AB)(t_1, t_2), & t_1 > t_2 \\ (BA)(t_2, t_1), & t_1 < t_2 \end{cases}$$

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$$\mathscr{T}\{\prod_{i=1}^{n} \sigma_i(t_i)\} = f(\sigma_1, \sigma_2, \ldots, \sigma_n)$$

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- and $\theta(t)$ is the Heaviside step function.

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- For our purposes, we can more generally just consider operators to be functions between vector spaces.

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  - $\mathcal{D}_1^n$ and $\mathcal{D}_n$ are both vector spaces, and
  - $\mathscr{T}$ maps elemets from $\mathcal{D}_1^n$ to $\mathcal{D}_n$

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- $\mathscr{T}$ is not an element of $\mathcal{O}_{\mathcal{H}}$, the set of linear operators on the Hilbert space.
- It's also not a *superoperator*: $\Sigma : \mathcal{O}_{\mathcal{H}} \to \mathcal{O}_{\mathcal{H}}$

Further Classifications

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